Dancing Dino

Starter:
https://scratch.mit.edu/projects/1282230734/editor

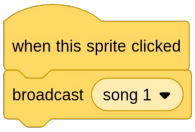
Don't forget to remix and rename.

1 In this project we will make the dinosaur dance! The starting sprites are;

- a dinosaur that has a collection of 'costumes' to make it animate
- a number that will be used as a button to start the music

2 First of all, in the Glow-1 sprite;

- broadcast a message 'song 1' when it is clicked
- We call the message 'song 1' because later we can add more songs later with other buttons



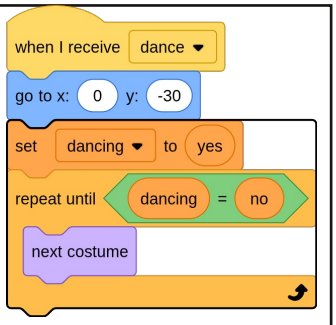
3 Next, in the **background**;

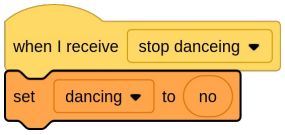
- Listen for the 'song 1' message
- Send the 'dance' message
- Play the music - make sure you use 'until done'
- Send the 'stop dancing' message

Note that the two messages we use here already exist so you do not need to create them.

4 Now in the **dinosaur** sprite;

- Create a new variable 'dancing' (make it for 'this sprite only')
- Set the 'dancing' variable to 'yes' when the sprite receives the message 'dance'
- Use a **repeat until** to keep changing to the next costume while the 'dancing' variable is 'yes'
- Set the 'dancing' variable to 'no' when the sprite receives the message 'stop dancing' - this will stop the dinosaur changing costumes





Why all the messages?

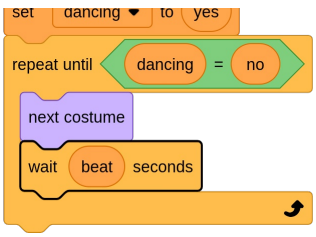
Each part - the button, the dinosaur and the background - **only** does it's own thing and then tells other parts to do their thing.

- The button **only** checks when it is pressed
- The background **only** plays the music
- The dinosaur **only** dances

This is called **separation of concerns** and it makes it easier to manage a program when it gets more complicated.

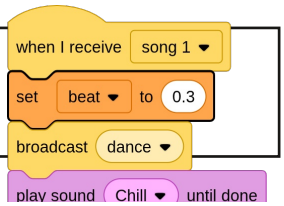
5 The dinosaur is dancing too wildly! We can use a variable to tell it how fast to move to fit the music. First, in the **dinosaur**;

- Create a new variable 'beat' (this must be 'for all sprites')
- Put a wait in the dancing loop for the length of time of 'beat'



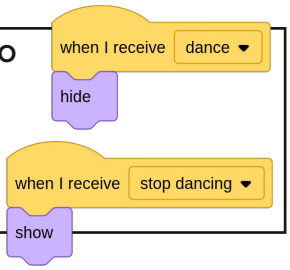
6 Now in the **background**;

- Set the value of the 'beat' variable to fit the music
- This might take some experimentation!



7 When the button is pressed it should disappear until it is ready to be pressed again. The button can do this by listening for the 'dance' and 'stop dancing' messages.

- Hide when the 'dance' message is received
- Show when the 'stop dancing' message is received



8 You may have been wondering why the button is a number '1'. This is because you can now add more buttons to play different music. You will need to;

- Create new buttons - the sprites of these numbers are all called 'Glow'
- Each new button needs all the scripts of the first button adjusted to appear in different places and send different messages
- The background needs new music - look in the 'Loops' section of the sounds
- For each button and each sound the background needs a new script to listen for the message and play the right music
- Each song may need a different 'beat' value - set the 'beat' variable in each script and try different values until the dinosaur dances in time

9 What else can you add? Here are some ideas;

- There could be another dinosaur dancing instead (or as well)
- Each piece of music could have a different background
- The dinosaur could change how it dances when you press different keys
- Something else from your own imagination!

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